
Order it
Ask Gemini
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Ask Google
Agent observes the current State
Agent selects an Action
Environment responds
Agent receives Reward
Agent moves to the Next State
Policy is updated through learning
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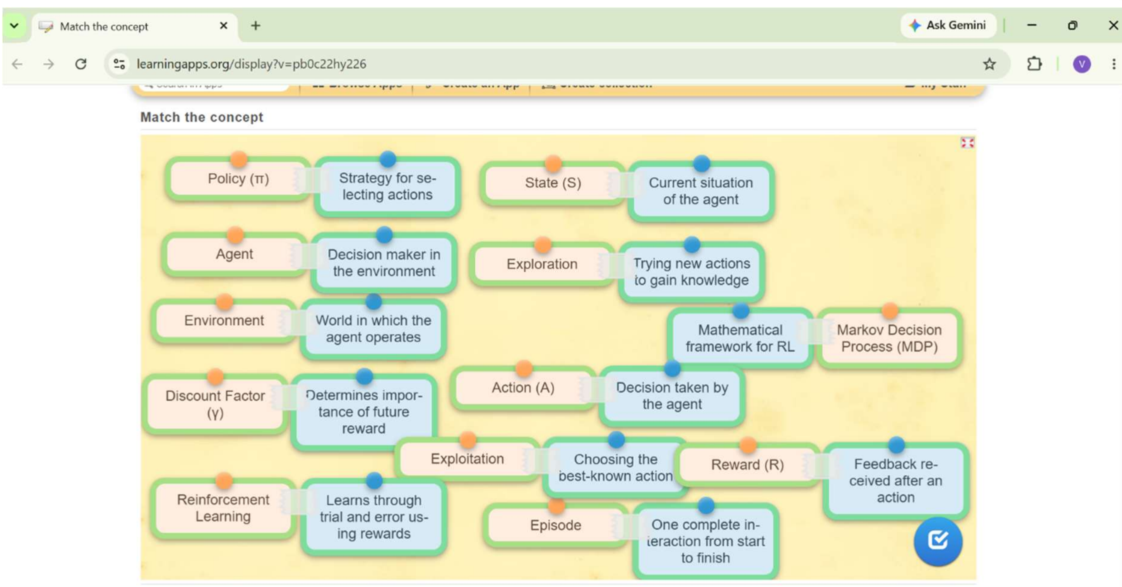
Match the concept
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Match the concept
Policy (π)
Strategy for selecting actions
State (S)
Current situation of the agent
Agent
Decision maker in the environment
Exploration
Trying new actions to gain knowledge
Environment
World in which the agent operates
Mathematical framework for RL
Markov Decision Process (MDP)
Discount Factor (γ)
Determines importance of future reward
Action (A)
Decision taken by the agent
Exploitation
Choosing the best-known action
Reward (R)
Feedback received after an action
Reinforcement Learning
Learns through trial and error using rewards
Episode
One complete interaction from start to finish

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Find the missing concepts

Reinforcement Learning is a branch of [Machine Learning] in which an [Agent] learns by interacting with an [Environment]. Instead of learning from labeled data, the agent improves its performance through trial and error by receiving [Rewards] after every action. The agent first observes the current [State], selects an appropriate [Action], and then moves to a new state based on the response of the environment. The strategy followed by the agent is known as a [Policy], whose objective is to maximize the cumulative [Reward] over multiple interactions. The mathematical framework used to model this interaction is called a [Markov Decision Process], which consists of [States], actions, rewards, and transition probabilities. During training, the agent balances [Exploration] by trying new actions and [Exploitation] by selecting actions that have produced higher rewards in the past. Reinforcement Learning is widely used in robotics, game playing, autonomous vehicles, and intelligent decision-making systems. Most Reinforcement Learning applications are developed using [Python] programming language. Popular libraries include [TensorFlow] for deep learning, [Keras] as a high-level neural network API, [NumPy] for numerical computations, and [Gymnasium] for providing standard environments used to train and evaluate RL agents. As the number of learning [Policy] increases, the agent gradually improves its [Policy] and makes better [Decisions] to achieve higher long-term rewards.

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Match the concept | Find the missing concepts | Words concept mapping | Untitled | Ask Gemini

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Untitled

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Words concept mapping

O	W	D	S	P	X	E	T	H	E	Y	K	J	C	E	E	N
D	H	A	I	B	Q	X	G	F	A	Y	S	M	B	P	L	U
M	Q	L	E	Z	X	Z	D	N	L	E	A	R	N	I	N	G
K	V	E	H	R	T	J	I	X	G	U	G	K	U	S	T	T
X	U	D	U	I	V	D	P	P	O	L	I	C	Y	O	R	I
V	T	F	I	F	P	M	S	J	R	X	F	J	J	D	A	N
T	D	R	L	N	D	E	N	V	I	R	O	N	M	E	N	T
A	Z	J	C	U	M	U	L	A	T	I	V	E	I	J	S	E
F	U	Y	P	L	F	M	A	C	H	I	N	E	I	V	I	R
R	E	I	N	F	O	R	C	E	M	E	N	T	H	U	T	A
K	U	U	X	P	T	K	N	O	H	A	O	H	M	L	I	C
A	W	D	Y	J	J	K	U	J	C	B	I	N	K	K	O	T
A	U	X	K	R	G	V	O	R	K	E	A	I	A	O	N	I
S	Q	L	L	O	A	X	I	Z	G	B	E	E	B	D	N	O
H	E	A	B	Z	B	O	C	I	B	U	P	T	G	L	V	N

1. CUMULATIVE
2. REINFORCEMENT
3. MACHINE
4. LEARNING
5. ENVIRONMENT
6. ALGORITHM
7. POLICY
8. TRANSITION
9. INTERACTION
10. EPISODE

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